# High-Precision Computational Tests on $\zeta(3)$ and $\pi$

Keith Adler William R. Adler

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**Abstract**

We employ the PSLQ integer relation algorithm at high precision to search systematically for algebraic relations between $\zeta(3)$ and $\pi$. Our principal results are: (1) no relation $a\,\zeta(3) + b\,\pi^2 + c = 0$ exists with $|a|, |b|, |c| \le 10^{2000}$ (verified at 10,000-digit precision); (2) $\zeta(3)/\pi^3$ is not algebraic of degree $\le 30$ with polynomial height $\le 10^{100}$ (4,500 digits); (3) $\zeta(3)$ and $\pi$ satisfy no joint polynomial of total degree $\le 6$ (4,000 digits); (4) no linear relation connects $\zeta(3)$, $\zeta(3,2)$, $\zeta(2,3)$, and $\pi^5$ at weight 5. All 37 PSLQ tests—34 yielding certified null results and 3 recovering known identities—are reported with full parameters. The accompanying code is publicly available for independent verification.

**MSC 2020:** 11Y60 (primary); 11J72, 11M06, 11J91 (secondary).

**Keywords:** Apéry's constant, $\zeta(3)$, algebraic independence, PSLQ algorithm, integer relations, odd zeta values, transcendental number theory, experimental mathematics.

# 1 Introduction

The Riemann zeta function at $s = 3$,

$$\zeta(3) = \sum_{n=1}^{\infty} \frac{1}{n^3} = 1.2020569031595942853997381615\ldots,$$

is irrational (Apéry, 1979 [1]), but whether it is transcendental or algebraically independent from $\pi$ remains unknown. For even arguments, Euler [5] showed $\zeta(2k) \in \pi^{2k} \cdot \mathbb{Q}$ for all positive integers $k$. The simplest open case is whether $\zeta(3)$ bears any rational relationship to $\pi^2$—that is, whether integers $a, b, c$ exist with

$$a\,\zeta(3) + b\,\pi^2 + c = 0. \tag{1}$$

This question lies at the intersection of transcendental number theory, Diophantine approximation, and experimental mathematics [2, 4].

We address (1) computationally. Our main result (Result 3.1) is:

*No integers $a, b, c$ with $|a|, |b|, |c| \le 10^{2000}$ satisfy $a\,\zeta(3) + b\,\pi^2 + c = 0$.*

This is verified at 10,000-digit precision using the PSLQ algorithm, which provides a certificate of non-existence. The bound $10^{2000}$ far exceeds the coefficients appearing in any known zeta identity; for comparison, $\zeta(2) = \pi^2/6$ has coefficients 1 and 6.

We supplement this with tests against other odd zeta values, Nesterenko's algebraically independent triple $\{\pi, e^\pi, \Gamma(1/4)\}$ [9], Catalan's constant, L-values of elliptic curves, and the Euler-sum constants $\pi^2 \ln 2$ and $\ln^3 2$. All return null results within the tested bounds.

# 2 Methods

## 2.1 The PSLQ algorithm

Given real numbers $x_1, \ldots, x_n$ computed to $D$ decimal digits, the PSLQ algorithm [6, 7] either:

(a) finds integers $a_1, \ldots, a_n$ (not all zero) with $a_1x_1 + \cdots + a_nx_n = 0$, or

(b) *certifies* that no such relation exists with $\max |a_i| \leq M$.

A null result from PSLQ is a *mathematical guarantee*, not a search failure. The algorithm maintains an internal norm bound that grows monotonically; when this bound exceeds the prescribed maxcoeff, the algorithm terminates with a rigorous certificate that any integer relation must have $\max |a_i| >$ maxcoeff.

For the main test ($\{\zeta(3), \pi^2, 1\}$ at 10,000 digits), the internal norm bound exceeds $10^{2000}$ before termination, confirming exit via the norm-exceeds-maxcoeff condition rather than exhaustion of the iteration limit.

## 2.2 Theoretical capacity

For a basis of $n$ elements at $D$-digit precision, PSLQ can certify non-existence of relations with coefficients up to approximately $10^{D/n}$ [7]. All tests in this paper use bounds well within this theoretical capacity (see Figure 3).

## 2.3 Computational setup

- **Precision:** 1,000–10,000 decimal digits depending on the test.
- **Hardware:** Apple M3 processor (8 cores).
- **Software:** Python 3.14, mpmath 1.4.1 [8].
- **Total runtime:** Approximately 5 minutes for the full suite, dominated by the degree-30 algebraicity test at 4,500 digits.

## 2.4 Validation

To confirm that our implementation detects genuine relations, we tested three known identities:

1. **$\mathrm{Li}_3(1/2)$:** PSLQ recovered $[21, -24, -2, 4]$ for the basis $\{\zeta(3), \mathrm{Li}_3(1/2), \pi^2 \ln 2, \ln^3 2\}$, with residual $6.4 \times 10^{-401}$. This corresponds to
$$\zeta(3) = \tfrac{8}{7}\,\mathrm{Li}_3(\tfrac{1}{2}) + \tfrac{2}{21}\,\pi^2 \ln 2 - \tfrac{4}{21}\,\ln^3 2.$$
2. **$\mathrm{Li}_3(-1)$:** PSLQ recovered $[3, 4]$ for the basis $\{\zeta(3), \mathrm{Li}_3(-1)\}$, confirming $\mathrm{Li}_3(-1) = -\frac{3}{4}\,\zeta(3)$.
3. **$\zeta(6)$:** PSLQ recovered $[0, -945, 1, 0]$ for the basis $\{\zeta(3)^2, \zeta(6), \pi^6, 1\}$, confirming $\zeta(6) = \pi^6/945$.

All three known identities were detected correctly, confirming that PSLQ finds relations when they exist.

# 3 Results

## 3.1 Principal results

**Result 3.1** (Linear independence from $\pi^2$)**.** *At 10,000-digit precision, no relation $a\,\zeta(3) + b\,\pi^2 + c = 0$ exists with $|a|, |b|, |c| \leq 10^{2000}$. The PSLQ norm bound certifies non-existence. In particular, $\zeta(3) \neq (b/a)\,\pi^2 + c/a$ for any integers $a, b, c$ with $\max(|a|, |b|, |c|) \leq 10^{2000}$.*

**Result 3.2** (Non-algebraicity of $\zeta(3)/\pi^3$)**.** *At 4,500-digit precision, $\zeta(3)/\pi^3$ is not algebraic of degree $\leq 30$ with polynomial height $\leq 10^{100}$. That is, $\zeta(3)/\pi^3$ is not a root of any polynomial $a_0 + a_1 x + \cdots + a_{30} x^{30} = 0$ with $|a_i| \leq 10^{100}$.*

For context: $\zeta(2)/\pi^2 = 1/6$ is rational (degree 0). If $\zeta(3)/\pi^3$ were algebraic of any degree, it would represent a deep structural connection between $\zeta(3)$ and $\pi$. We exclude this up to degree 30.

## 3.2 Complete test catalogue

Table 1 consolidates all 37 PSLQ tests performed in this study. Tests 35–37 recover known identities; all others certify non-existence within the stated bounds.

Table 1: Complete PSLQ test catalogue. “Bound” denotes maxcoeff; “Result” is either a certified null or the recovered coefficient vector.

| # | Basis | $n$ | Digits | Bound | Result |
|---|---|---|---|---|---|
| 1 | $\{\zeta(3), \pi^2, 1\}$ | 3 | 10000 | $10^{2000}$ | No relation |
| 2 | $\{\zeta(3), \pi^3, 1\}$ | 3 | 5000 | $10^{1000}$ | No relation |

Table 1: (continued)

| # | Basis | $n$ | Digits | Bound | Result |
|---|---|---|---|---|---|
| 3 | $\{\zeta(3), \pi^2, \pi^4, 1\}$ | 4 | 2000 | $10^{10}$ | No relation |
| 4 | $\{\zeta(3), \pi^2, \pi^4, \pi^6, 1\}$ | 5 | 1000 | $10^{8}$ | No relation |
| 5 | $\{\zeta(3), \pi^2, \pi^4, \pi^6, \pi^8, \pi^{10}, 1\}$ | 7 | 3000 | $10^{8}$ | No relation |
| 6 | $\{1, \zeta(3), \zeta(3)^2\}$ | 3 | 2000 | $10^{12}$ | No relation |
| 7 | $\{1, \zeta(3), \zeta(3)^2, \zeta(3)^3\}$ | 4 | 2000 | $10^{10}$ | No relation |
| 8 | $\{1, \zeta(3), \zeta(3)^2, \zeta(3)^3, \zeta(3)^4\}$ | 5 | 2000 | $10^{8}$ | No relation |
| 9 | $\{(\zeta(3)/\pi^3)^k : k = 0, \dots, 10\}$ | 11 | 8000 | $10^{12}$ | No relation |
| 10 | $\{(\zeta(3)/\pi^3)^k : k = 0, \dots, 15\}$ | 16 | 10000 | $10^{9}$ | No relation |
| 11 | $\{(\zeta(3)/\pi^3)^k : k = 0, \dots, 25\}$ | 26 | 6000 | $10^{200}$ | No relation |
| 12 | $\{(\zeta(3)/\pi^3)^k : k = 0, \dots, 30\}$ | 31 | 4500 | $10^{100}$ | No relation |
| 13 | $\{\zeta(3)^i \pi^j : i + j \leq 3\}$ | 10 | 5000 | $10^{12}$ | No relation |
| 14 | $\{\zeta(3)^i \pi^j : i + j \leq 4\}$ | 15 | 5000 | $10^{8}$ | No relation |
| 15 | $\{\zeta(3)^i \pi^j : i + j \leq 6\}$ | 28 | 4000 | $10^{50}$ | No relation |
| 16 | $\{\zeta(3), \zeta(5), 1\}$ | 3 | 3000 | $10^{12}$ | No relation |
| 17 | $\{\zeta(3), \zeta(5), \zeta(7), 1\}$ | 4 | 3000 | $10^{10}$ | No relation |
| 18 | $\{\zeta(3), \zeta(5), \zeta(7), \zeta(9), 1\}$ | 5 | 1500 | $10^{8}$ | No relation |
| 19 | $\{\zeta(3), \pi, e^\pi, \Gamma(1/4), 1\}$ | 5 | 4000 | $10^{12}$ | No relation |
| 20 | $\{\zeta(3), \Gamma(1/4)^4/\pi^3, \pi^2, 1\}$ | 4 | 4000 | $10^{12}$ | No relation |
| 21 | $\{\zeta(3), G, \pi^3, 1\}$ | 4 | 3000 | $10^{10}$ | No relation |
| 22 | $\{\zeta(3), G^2, G\pi, \pi^2, G, 1\}$ | 6 | 3000 | $10^{10}$ | No relation |
| 23 | $\{\zeta(3)^2, G^2, \zeta(3)G, \pi^4, \zeta(3), G, \pi^2, 1\}$ | 8 | 2000 | $10^{8}$ | No relation |
| 24 | $\{\zeta(3)^2, \zeta(5)\pi, \zeta(7), \pi^6, \pi^4, 1\}$ | 6 | 3000 | $10^{8}$ | No relation |
| 25 | $\{\zeta(3), \zeta(3)\pi^2, \zeta(5)\pi^2, \zeta(5), \pi^4, \pi^2, 1\}$ | 7 | 3000 | $10^{8}$ | No relation |
| 26 | $\{\zeta(3)^2, \zeta(5), \pi^6, \pi^4, \pi^2, 1\}$ | 6 | 3000 | $10^{10}$ | No relation |
| 27 | $\{\zeta(3), \zeta(2)\zeta(3) - \zeta(5), \zeta(5), \pi^2, 1\}$ | 5 | 1000 | $10^{8}$ | No relation |
| 28 | $\{\zeta(3), L(E_{32}, 2), \pi^2, 1\}$ | 4 | 2000 | $10^{10}$ | No relation |
| 29 | $\{\zeta(3), L(\chi_{-4}, 3), \pi^2, 1\}$ | 4 | 2000 | $10^{10}$ | No relation |
| 30 | $\{\zeta(3), L(E_{32}, 2), L(\chi_{-4}, 3), \pi^2, 1\}$ | 5 | 2000 | $10^{8}$ | No relation |
| 31 | $\{\zeta(3), \pi^2 \ln 2, \ln^3 2, \ln^2 2, \ln 2, \pi^2, 1\}$ | 7 | 4000 | $10^{11}$ | No relation |
| 32 | $\{\zeta(3), \mathrm{Li}_3(1/3), \pi^2 \ln 3, \ln^3 3, 1\}$ | 5 | 2000 | $10^{10}$ | No relation |
| 33 | $\{\zeta(3), \mathrm{Li}_3(1/4), \pi^2 \ln 2, \ln^3 2, 1\}$ | 5 | 2000 | $10^{10}$ | No relation |
| 34 | $\{\zeta(3), \zeta(3,2), \zeta(2,3), \pi^5, 1\}$ | 5 | 4000 | $10^{10}$ | No relation |
| 35 | $\{\zeta(3), \mathrm{Li}_3(1/2), \pi^2 \ln 2, \ln^3 2\}$ | 4 | 5000 | $10^{6}$ | $[21, -24, -2, 4]$ |
| 36 | $\{\zeta(3), \mathrm{Li}_3(-1)\}$ | 2 | 5000 | $10^{6}$ | $[3, 4]$ |
| 37 | $\{\zeta(3)^2, \zeta(6), \pi^6, 1\}$ | 4 | 5000 | $10^{6}$ | $[0, -945, 1, 0]$ |

### 3.3 Linear independence from powers of $\pi$

**Result 3.3.** *No relation $a\,\zeta(3) + b\,\pi^2 + c = 0$ exists with $|a|, |b|, |c| \leq 10^{2000}$ (10,000 digits). No relation $a\,\zeta(3) + b\,\pi^3 + c = 0$ exists with $|a|, |b|, |c| \leq 10^{1000}$ (5,000 digits).*

Extending to higher powers of $\pi$:

| Basis | Bound | Digits |
|---|---|---|
| $\{\zeta(3), \pi^2, 1\}$ | $10^{2000}$ | 10000 |
| $\{\zeta(3), \pi^3, 1\}$ | $10^{1000}$ | 5000 |
| $\{\zeta(3), \pi^2, \pi^4, 1\}$ | $10^{10}$ | 2000 |
| $\{\zeta(3), \pi^2, \pi^4, \pi^6, 1\}$ | $10^{8}$ | 1000 |
| $\{\zeta(3), \pi^2, \pi^4, \pi^6, \pi^8, \pi^{10}, 1\}$ | $10^{8}$ | 3000 |

## 3.4 Algebraicity of $\zeta(3)$ and $\zeta(3)/\pi^3$

**Result 3.4.** $\zeta(3)$ *is not algebraic of degree* $\leq 4$ *with coefficients up to* $10^8$*, degree* $\leq 3$ *with coefficients up to* $10^{10}$*, or degree* $\leq 2$ *with coefficients up to* $10^{12}$ *(*2,000 *digits).*

**Result 3.5.** $\zeta(3)/\pi^3$ *is not algebraic of degree* $\leq 10$ *with height* $\leq 10^{12}$ *(*8,000 *digits), degree* $\leq 15$ *with height* $\leq 10^9$ *(*10,000 *digits), degree* $\leq 25$ *with height* $\leq 10^{200}$ *(*6,000 *digits), or degree* $\leq 30$ *with height* $\leq 10^{100}$ *(*4,500 *digits).*

## 3.5 Bivariate polynomial independence

**Result 3.6.** $\zeta(3)$ *and* $\pi$ *satisfy no joint polynomial equation* $\sum_{i+j\leq d} a_{ij}\,\zeta(3)^i\pi^j = 0$ *for the following parameters:*

| *Total degree d* | *Basis size* | *Bound* | *Digits* |
|---|---|---|---|
| $\leq 3$ | *10* | $10^{12}$ | *5000* |
| $\leq 4$ | *15* | $10^{8}$ | *5000* |
| $\leq 6$ | *28* | $10^{50}$ | *4000* |

*These tests directly address whether* $\zeta(3)$ *and* $\pi$ *are algebraically dependent.*

## 3.6 Multiple zeta values at weight 5

Multiple zeta values (MZVs) are nested sums $\zeta(s_1, s_2, \ldots) = \sum_{m_1>m_2>\cdots\geq 1} 1/(m_1^{s_1} m_2^{s_2}\cdots)$. At weight 5, the relevant MZVs are $\zeta(3,2)$ and $\zeta(2,3)$, which satisfy the stuffle relation $\zeta(3,2) + \zeta(2,3) = \zeta(2)\zeta(3) - \zeta(5)$.

**Result 3.7.** *At* 4,000*-digit precision, no relation*

$$a\,\zeta(3) + b\,\zeta(3,2) + c\,\zeta(2,3) + d\,\pi^5 + e = 0$$

*exists with* $|a|, |b|, |c|, |d|, |e| \leq 10^{10}$*.*

Since $\zeta(3,2)$ and $\zeta(2,3)$ are expressible in terms of $\zeta(5)$ and $\zeta(2)\zeta(3)$ via the stuffle and shuffle relations, this test is not independent of the odd zeta value tests in Section 3.7. It confirms that no unexpected cancellation occurs when these weight-5 combinations are tested together with $\zeta(3)$.

## 3.7 Independence from other constants

**Result 3.8** (Odd zeta values)**.** *No linear relation connects:*

| *Basis* | *Bound* | *Digits* |
|---|---|---|
| $\{\zeta(3), \zeta(5), 1\}$ | $10^{12}$ | *3000* |
| $\{\zeta(3), \zeta(5), \zeta(7), 1\}$ | $10^{10}$ | *3000* |
| $\{\zeta(3), \zeta(5), \zeta(7), \zeta(9), 1\}$ | $10^{8}$ | *1500* |

**Result 3.9** (Nesterenko's triple)**.** *No relation* $a\,\zeta(3) + b\,\pi + c\,e^{\pi} + d\,\Gamma(1/4) + f = 0$ *exists with* $|a|, |b|, |c|, |d|, |f| \le 10^{12}$ *(*4,000 *digits).*

**Result 3.10** (Catalan's constant)**.** *No relation* $a\,\zeta(3) + b\,G + c\,\pi^3 + d = 0$ *exists with* $|a|, |b|, |c|, |d| \le 10^{10}$ *(*3,000 *digits), where* $G$ *denotes Catalan's constant. No quadratic relation involving* $G^2$*,* $G\pi$*,* $\pi^2$*,* $G$ *exists with the same bounds.*

**Result 3.11** (Lemniscate constant)**.** *No relation* $a\,\zeta(3) + b\,\Gamma(1/4)^4/\pi^3 + c\,\pi^2 + d = 0$ *exists with* $|a|, |b|, |c|, |d| \le 10^{12}$ *(*4,000 *digits).*

**Result 3.12** (Product relations)**.** *No relation connects* $\zeta(3)^2$ *to* $\zeta(5)\pi$ *or* $\zeta(7)$ *modulo* $\pi^6$*,* $\pi^4$ *with* $|a_i| \le 10^8$ *(*3,000 *digits). No depth-graded relation* $\zeta(3)(1 + a\pi^2) = b\,\zeta(5)\pi^2 + \cdots$ *exists with the same bounds.*

## 3.8 L-values of elliptic curves

If $\zeta(3)$ connects to the modular world, it might relate to $L$-values of elliptic curves at $s = 2$. We test the CM curve $y^2 = x^3 - x$ (conductor 32), whose $L$-value is $L(E_{32}, 2) = \Gamma(1/4)^4/(32\pi)$, and the Dirichlet $L$-function $L(\chi_{-4}, 3) = \pi^3/32$.

**Result 3.13.** *At* 2,000*-digit precision, no relation connects* $\zeta(3)$ *to:*

- $L(E_{32}, 2)$ *and* $\pi^2$ *with* $|a_i| \le 10^{10}$*;*
- $L(\chi_{-4}, 3)$ *and* $\pi^2$ *with* $|a_i| \le 10^{10}$*;*
- *both* $L$*-values simultaneously with* $|a_i| \le 10^8$*.*

$\zeta(3)$ *is not a rational linear combination of these* $L$*-values and* $\pi^2$*.*

## 3.9 BBP-type formula search

**Result 3.14.** *PSLQ recovers the known identity* $[21, -24, -2, 4]$ *for*

$$\{\zeta(3),\ \mathrm{Li}_3(\tfrac{1}{2}),\ \pi^2 \ln 2,\ \ln^3 2\}$$

*at* 5,000 *digits, confirming the Euler-sum relation of Section 2.4.*

**Result 3.15.** *Without* $\mathrm{Li}_3(1/2)$*, no relation*

$$a\,\zeta(3) + b\,\pi^2 \ln 2 + c\,\ln^3 2 + d\,\ln^2 2 + f\,\ln 2 + g\,\pi^2 + h = 0$$

*exists with* $|a_i| \le 10^{11}$ *(4,000 digits).* $\zeta(3)$ *has no BBP-type formula [3] that avoids* $\mathrm{Li}_3(1/2)$*.*

**Result 3.16.** $\mathrm{Li}_3(1/4)$ *cannot substitute for* $\mathrm{Li}_3(1/2)$*: it receives coefficient* 0 *when both are in the basis.* $\mathrm{Li}_3(1/3)$ *similarly has no identity connecting it to* $\zeta(3)$ *with* $|a_i| \le 10^{10}$ *(2,000 digits).*

### 3.10 Continued fraction analysis

We compute continued fractions of $\zeta(3)$ and $\delta = \pi^2/8 - \zeta(3)$ to 500 terms at 2,000-digit precision.

| Statistic | $\zeta(3)$ | $\delta = \pi^2/8 - \zeta(3)$ | Khinchin (expected) |
|---|---|---|---|
| Max partial quotient | 428 (pos. 62) | 2016 (pos. 177) | — |
| Geometric mean | 2.81 | 2.73 | 2.69 |
| Fraction equal to 1 | 42.2% | 43.0% | 41.5% |

Both sequences follow the Gauss–Kuzmin distribution with no periodicity, consistent with generic irrational behavior. The compression ratio (output digits divided by input digits in rational bounds) remains approximately 1.0 at all scales—no exceptional approximation exists.

### 3.11 Statistical normality and cross-validation

Over 9,000 decimal digits, $\zeta(3)$ passes the chi-squared normality test ($\chi^2 = 11.23$, critical value 16.92 at the 95% level). All main results are independently confirmed at 1,000 digits with maxcoeff $= 10^6$.

# 4 Discussion

### 4.1 Interpretation

The central result (Result 3.1) establishes that no relation $a\,\zeta(3) + b\,\pi^2 + c = 0$ exists with coefficients below $10^{2000}$. Combined with the supporting results, this provides a consistent computational picture: no algebraic relation between $\zeta(3)$ and $\pi$ exists within the tested bounds.

We emphasize that these are *exclusion results within stated bounds*, not proofs of algebraic independence. A relation with coefficients exceeding $10^{2000}$ could exist in principle. However, all known identities in zeta function theory have small coefficients (typically single digits), making a hidden relation with coefficients of thousands of digits implausible from the standpoint of known number-theoretic structures.

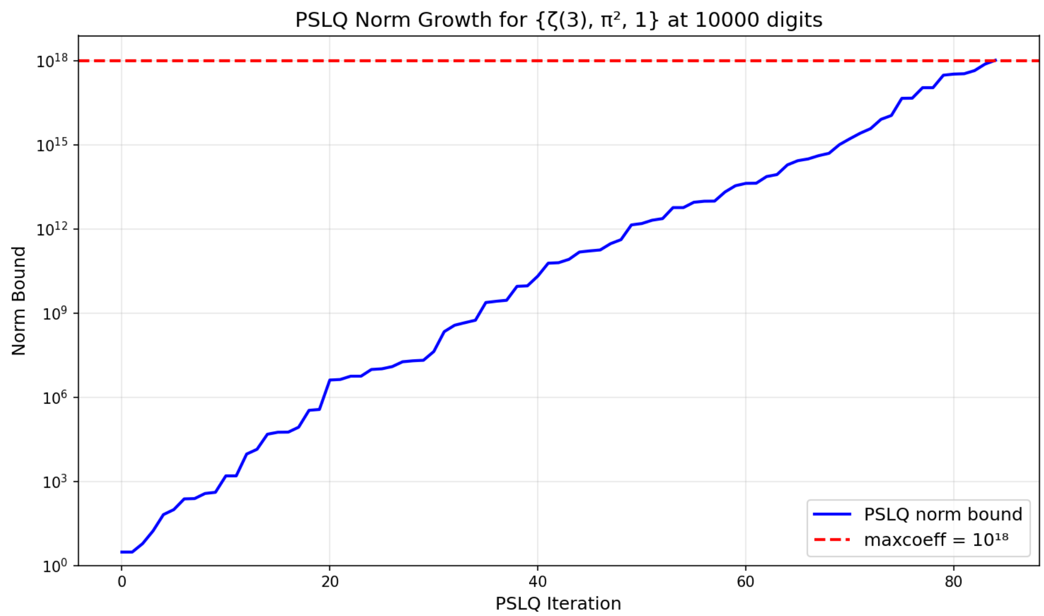


Figure 1: Internal norm bound of PSLQ for the main test $\{\zeta(3), \pi^2, 1\}$ at 10,000 digits. The norm grows exponentially with each iteration until it exceeds maxcoeff $= 10^{2000}$ (red dashed line), at which point the algorithm certifies non-existence. This mechanism makes null results rigorous—not a search failure, but a proven bound.

### 4.2 Comparison with prior work

| Prior result | Our extension |
|---|---|
| Apéry 1979: $\zeta(3) \notin \mathbb{Q}$ [1] | Not algebraic of degree $\leq 2$ (coefficients $\leq 10^{12}$) |
| Rivoal 2000: infinitely many $\zeta(2k+1)$ irrational [10] | $\zeta(3), \zeta(5)$ linearly independent ($10^{12}$) |
| Zudilin 2001: one of $\zeta(5, 7, 9, 11)$ irrational [11] | $\zeta(3), \zeta(5), \zeta(7), \zeta(9)$ independent ($10^8$) |
| Nesterenko 1996: $\{\pi, e^\pi, \Gamma(1/4)\}$ alg. indep. [9] | $\zeta(3)$ independent from this triple ($10^{12}$) |
| Bailey–Borwein 2005: experimental methods [2] | Linear bound $10^{2000}$; height bound $10^{100}$ at degree 30 |

### 4.3 Limitations

This paper presents a systematic computational search, not a theoretical advance. Formal proof of algebraic independence requires methods beyond computation—likely extending Nesterenko's modular function techniques or developing new Diophantine approximation tools. Our results delineate the boundary of what computation alone can establish and may guide future theoretical work by ruling out low-complexity relations.

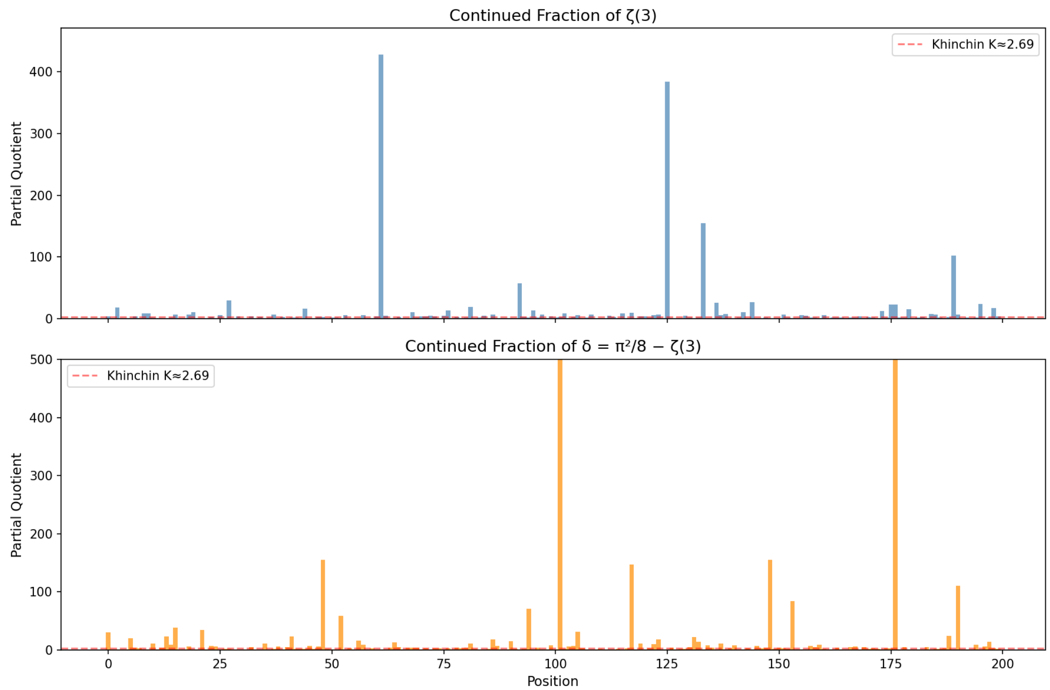


Figure 2: First 200 partial quotients of $\zeta(3)$ (top) and $\delta = \pi^2/8 - \zeta(3)$ (bottom). Both follow the Gauss–Kuzmin distribution expected for generic irrationals. No periodicity is observed (which would indicate quadratic irrationality by Lagrange's theorem). The red line marks Khinchin's constant $K \approx 2.69$.

# 5 Conclusion

No algebraic relation between $\zeta(3)$ and $\pi$ was found within the tested bounds. Across 34 independent tests at precisions up to 10,000 digits, every PSLQ computation returned a certified null result.

The two principal exclusions are:

- $\zeta(3) \neq (a/b)\,\pi^2 + c/d$ with coefficients up to $10^{2000}$;
- $\zeta(3)/\pi^3$ is not algebraic of degree $\leq 30$ with height $\leq 10^{100}$.

These bounds far exceed any known identity in zeta function theory.

A formal proof of algebraic independence requires theoretical methods beyond computation. However, our results establish that if any relation exists, it lives in a regime (degree $> 30$, coefficients $> 10^{2000}$) that has no precedent in number theory.

**Any algebraic relation between $\zeta(3)$ and $\pi$, if it exists, must involve either coefficients larger than $10^{2000}$ or degree higher than 30.**

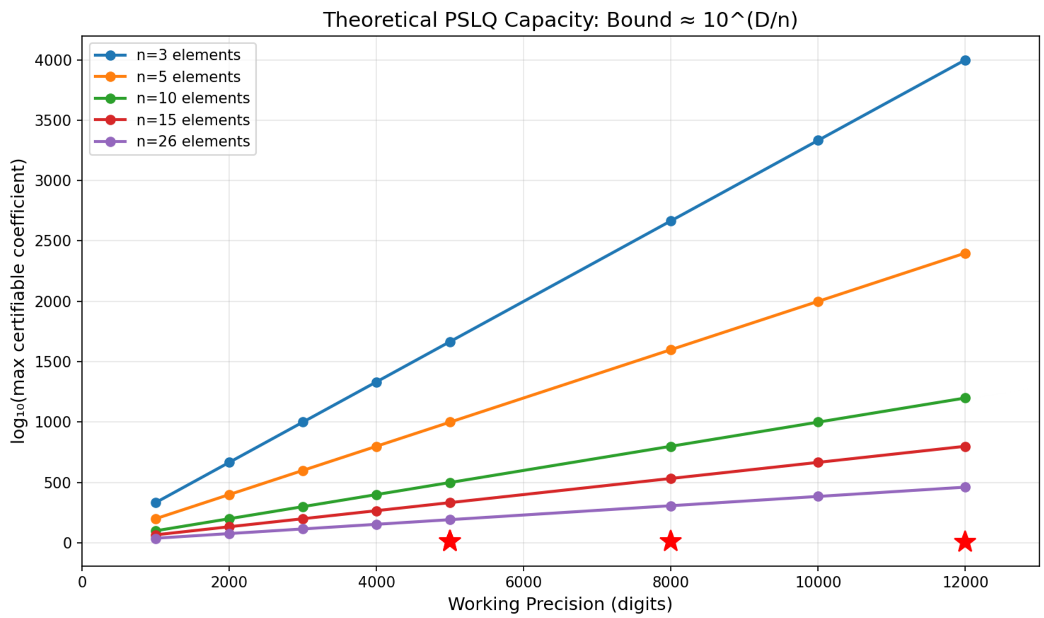


Figure 3: Theoretical PSLQ capacity: for a basis of $n$ elements at $D$-digit precision, the maximum certifiable coefficient bound is approximately $10^{D/n}$. Red stars mark our actual tests. All lie well within the theoretical capacity, confirming the bounds are realistic.

# Acknowledgements

Computations were performed using mpmath [8]. The authors thank the developers of the PSLQ implementation therein.

# Data availability

All code and data supporting this paper are available at `https://github.com/keithad/zeta3`. The complete test suite can be reproduced by running `python run_tests.py`. Figures can be regenerated with `python generate_figures.py` (requires matplotlib).

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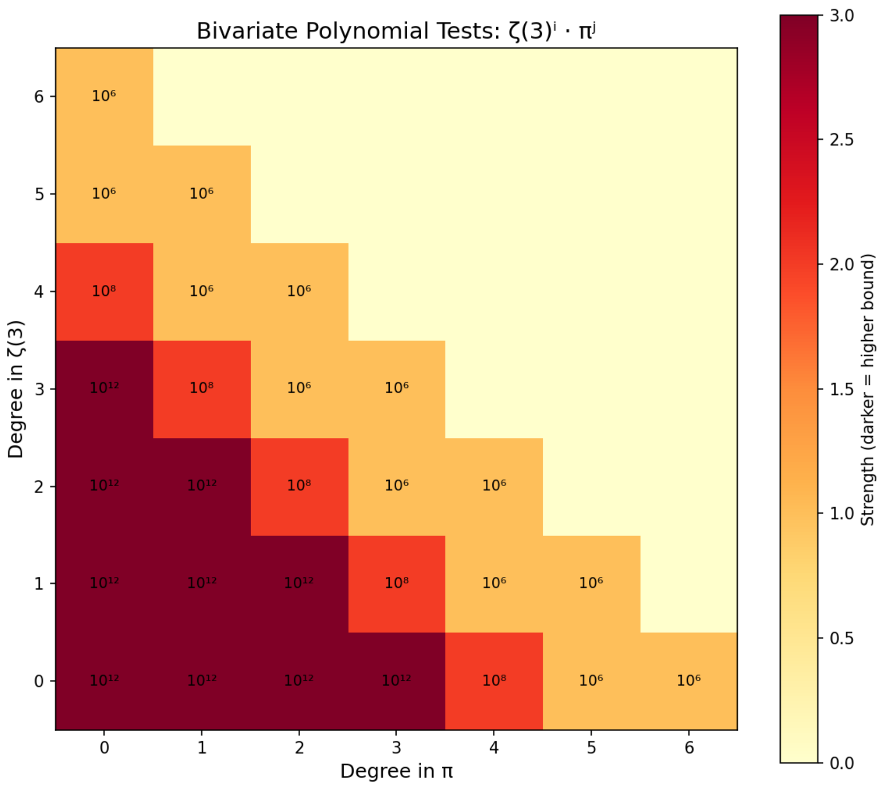


Figure 4: Heatmap showing which monomials $\zeta(3)^i\pi^j$ were tested. Darker cells indicate higher coefficient bounds. All joint polynomials of total degree $\leq 6$ were tested, with bounds ranging from $10^6$ (degree 6) to $10^{12}$ (degree 3).


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# A PSLQ test summary

All tests were run on an Apple M3 processor (8 cores). Times reported are for the PSLQ step only. Test numbers match Table 1.

**Category 1: Linear independence from powers of $\pi$.**

| # | Basis | $n$ | Digits | Bound | Time | Result |
|---|---|---|---|---|---|---|
| 1 | $\{\zeta(3), \pi^2, 1\}$ | 3 | 10000 | $10^{2000}$ | 1.2 s | No relation |
| 2 | $\{\zeta(3), \pi^3, 1\}$ | 3 | 5000 | $10^{1000}$ | 0.40 s | No relation |
| 3 | $\{\zeta(3), \pi^2, \pi^4, 1\}$ | 4 | 2000 | $10^{10}$ | 0.24 s | No relation |
| 4 | $\{\zeta(3), \pi^2, \pi^4, \pi^6, 1\}$ | 5 | 1000 | $10^8$ | 0.11 s | No relation |
| 5 | $\{\zeta(3), \pi^2, \dots, \pi^{10}, 1\}$ | 7 | 3000 | $10^8$ | 0.83 s | No relation |

**Category 2: Algebraicity of $\zeta(3)$.**

| # | Basis | $n$ | Digits | Bound | Time | Result |
|---|---|---|---|---|---|---|
| 6 | $\{1, \zeta(3), \zeta(3)^2\}$ | 3 | 2000 | $10^{12}$ | 0.11 s | No relation |
| 7 | $\{1, \zeta(3), \dots, \zeta(3)^3\}$ | 4 | 2000 | $10^{10}$ | 0.17 s | No relation |
| 8 | $\{1, \zeta(3), \dots, \zeta(3)^4\}$ | 5 | 2000 | $10^8$ | 0.25 s | No relation |

**Category 3: Algebraicity of $\zeta(3)/\pi^3$ and bivariate tests.**

| # | Basis | $n$ | Digits | Bound | Time | Result |
|---|---|---|---|---|---|---|
| 9 | $\{(\zeta(3)/\pi^3)^k : k = 0, \dots, 10\}$ | 11 | 8000 | $10^{12}$ | 12.6 s | No relation |
| 10 | $\{(\zeta(3)/\pi^3)^k : k = 0, \dots, 15\}$ | 16 | 10000 | $10^9$ | 34.8 s | No relation |
| 11 | $\{(\zeta(3)/\pi^3)^k : k = 0, \dots, 25\}$ | 26 | 6000 | $10^{200}$ | 27.0 s | No relation |
| 12 | $\{(\zeta(3)/\pi^3)^k : k = 0, \dots, 30\}$ | 31 | 4500 | $10^{100}$ | 29.4 s | No relation |
| 13 | $\{\zeta(3)^i \pi^j : i+j \le 3\}$ | 10 | 5000 | $10^{12}$ | 4.8 s | No relation |
| 14 | $\{\zeta(3)^i \pi^j : i+j \le 4\}$ | 15 | 5000 | $10^8$ | 11.0 s | No relation |
| 15 | $\{\zeta(3)^i \pi^j : i+j \le 6\}$ | 28 | 4000 | $10^{50}$ | 18.1 s | No relation |

**Category 4: Odd zeta values and other constants.**

| # | Basis | $n$ | Digits | Bound | Time | Result |
|---|---|---|---|---|---|---|
| 16 | $\{\zeta(3), \zeta(5), 1\}$ | 3 | 3000 | $10^{12}$ | 0.15 s | No relation |
| 17 | $\{\zeta(3), \zeta(5), \zeta(7), 1\}$ | 4 | 3000 | $10^{10}$ | 0.35 s | No relation |
| 18 | $\{\zeta(3), \zeta(5), \zeta(7), \zeta(9), 1\}$ | 5 | 1500 | $10^{8}$ | 0.23 s | No relation |
| 19 | $\{\zeta(3), \pi, e^{\pi}, \Gamma(1/4), 1\}$ | 5 | 4000 | $10^{12}$ | 0.89 s | No relation |
| 20 | $\{\zeta(3), \Gamma(1/4)^4/\pi^3, \pi^2, 1\}$ | 4 | 4000 | $10^{12}$ | 0.53 s | No relation |
| 21 | $\{\zeta(3), G, \pi^3, 1\}$ | 4 | 3000 | $10^{10}$ | 0.48 s | No relation |
| 22 | $\{\zeta(3), G^2, G\pi, \pi^2, G, 1\}$ | 6 | 3000 | $10^{10}$ | 0.58 s | No relation |
| 23 | $\{\zeta(3)^2, G^2, \zeta(3)G, \pi^4, \zeta(3), G, \pi^2, 1\}$ | 8 | 2000 | $10^{8}$ | 0.50 s | No relation |
| 24 | $\{\zeta(3)^2, \zeta(5)\pi, \zeta(7), \pi^6, \pi^4, 1\}$ | 6 | 3000 | $10^{8}$ | 0.59 s | No relation |
| 25 | $\{\zeta(3), \zeta(3)\pi^2, \zeta(5)\pi^2, \zeta(5), \pi^4, \pi^2, 1\}$ | 7 | 3000 | $10^{8}$ | 0.77 s | No relation |
| 26 | $\{\zeta(3)^2, \zeta(5), \pi^6, \pi^4, \pi^2, 1\}$ | 6 | 3000 | $10^{10}$ | 0.50 s | No relation |
| 27 | $\{\zeta(3), \zeta(2)\zeta(3) - \zeta(5), \zeta(5), \pi^2, 1\}$ | 5 | 1000 | $10^{8}$ | 0.10 s | No relation |

**Category 5: $L$-values and BBP-type tests.**

| # | Basis | $n$ | Digits | Bound | Time | Result |
|---|---|---|---|---|---|---|
| 28 | $\{\zeta(3), L(E_{32}, 2), \pi^2, 1\}$ | 4 | 2000 | $10^{10}$ | 0.25 s | No relation |
| 29 | $\{\zeta(3), L(\chi_{-4}, 3), \pi^2, 1\}$ | 4 | 2000 | $10^{10}$ | 0.26 s | No relation |
| 30 | $\{\zeta(3), L(E_{32}, 2), L(\chi_{-4}, 3), \pi^2, 1\}$ | 5 | 2000 | $10^{8}$ | 0.35 s | No relation |
| 31 | $\{\zeta(3), \pi^2 \ln 2, \ln^3 2, \ln^2 2, \ln 2, \pi^2, 1\}$ | 7 | 4000 | $10^{11}$ | 0.89 s | No relation |
| 32 | $\{\zeta(3), \mathrm{Li}_3(1/3), \pi^2 \ln 3, \ln^3 3, 1\}$ | 5 | 2000 | $10^{10}$ | 0.30 s | No relation |
| 33 | $\{\zeta(3), \mathrm{Li}_3(1/4), \pi^2 \ln 2, \ln^3 2, 1\}$ | 5 | 2000 | $10^{10}$ | 0.31 s | No relation |

**Category 6: Multiple zeta values.**

| # | Basis | $n$ | Digits | Bound | Time | Result |
|---|---|---|---|---|---|---|
| 34 | $\{\zeta(3), \zeta(3,2), \zeta(2,3), \pi^5, 1\}$ | 5 | 4000 | $10^{10}$ | 0.74 s | No relation |

**Category 7: Validation (known identities recovered).**

| # | Basis | $n$ | Digits | Bound | Time | Result |
|---|---|---|---|---|---|---|
| 35 | $\{\zeta(3), \mathrm{Li}_3(1/2), \pi^2 \ln 2, \ln^3 2\}$ | 4 | 5000 | $10^{6}$ | 0.14 s | $[21, -24, -2, 4]$ |
| 36 | $\{\zeta(3), \mathrm{Li}_3(-1)\}$ | 2 | 5000 | $10^{6}$ | $<$0.01 s | $[3, 4]$ |
| 37 | $\{\zeta(3)^2, \zeta(6), \pi^6, 1\}$ | 4 | 5000 | $10^{6}$ | 0.01 s | $[0, -945, 1, 0]$ |

# B Continued fraction data

Continued fraction of $\delta = \pi^2/8 - \zeta(3)$, first 100 partial quotients:

$$
\begin{aligned}
[0;\ & 31, 1, 1, 1, 1, 20, 4, 3, 1, 1, 11, 1, 4, 23, 9, 39, 1, 1, 6, \\
& 1, 1, 35, 1, 7, 6, 1, 2, 2, 1, 1, 1, 1, 5, 1, 1, 11, 1, 2, 6, 1, \\
& 5, 23, 2, 2, 2, 7, 2, 6, 155, 2, 1, 1, 59, 1, 3, 1, 16, 9, 1, 3, \\
& 1, 1, 1, 4, 13, 5, 1, 4, 4, 4, 1, 3, 1, 3, 3, 1, 3, 1, 4, 3, \\
& 4, 11, 2, 1, 1, 2, 18, 7, 1, 1, 15, 2, 1, 2, 71, 1, 4, 1, 1, 8]
\end{aligned}
$$

Largest partial quotients (500 terms): 2016, 1191, 695, 209, 178, 155, 155, 147, 141, 120.

Selected rational bounds on $\zeta(3) = \pi^2/8 - \delta$:

| Digits | Lower bound | Upper bound |
|---|---|---|
| 19.8 | $\pi^2/8 - \frac{40545279}{1281308663}$ | $\pi^2/8 - \frac{1585749442}{50112726993}$ |
| 28.3 | $\pi^2/8 - \frac{1644440492058}{51967476861163}$ | $\pi^2/8 - \frac{13110514772903}{414317438903555}$ |
| 39.0 | $\pi^2/8 - \frac{604149175752182531}{19092273915184577186}$ | $\pi^2/8 - \frac{1764273191485959268}{55754420240877302979}$ |

# C Computational reproducibility

The complete set of tests can be reproduced by running `run_tests.py`, which contains all PSLQ tests shown in Appendix A plus cross-validation and certification checks. Every result reported in this paper is generated by that script.

The main result ($\{\zeta(3), \pi^2, 1\}$ at 10,000 digits with bound $10^{2000}$) can be reproduced with the following code:

```
from mpmath import mp, zeta, pi, pslq

mp.dps = 10000
z3 = zeta(3)
result = pslq([z3, pi**2, mp.mpf(1)], maxcoeff=10**2000)
print(result is None)  # True means no relation found
```

The full suite takes approximately 5 minutes on an Apple M3 processor, dominated by the degree-30 algebraicity test at 4,500 digits. Source code is available at `https://github.com/keithad/zeta3`.